

# Ewan Townshend

 etown.dev

 ewantown

 ewan-townshend

*Polyglot developer with experience building and integrating high-volume claims management systems. Comfortable in Java, TypeScript, C#, SQLs and Lisps. Adept at API and system design. Prefers functional and symbolic programming and metaprogramming. Interested in programming language theory, combinatorial search problems, and (neuro)symbolic AI.*

## Employment

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| Accenture                                                                                                                                                                                                                                                                                                                                | VANCOUVER, B.C.   |
| <b>Software Engineering Specialist</b>                                                                                                                                                                                                                                                                                                   | 08/2025 – present |
| <ul style="list-style-type: none"><li>• Consulting role with a focus on designing and building high-volume claims management systems.</li><li>• Led design and build of APIs, batch jobs, and a telemetry system supporting security audits.</li><li>• Work primarily on enterprise Java projects based on the Cúram platform.</li></ul> |                   |
| WorkSafeBC                                                                                                                                                                                                                                                                                                                               | RICHMOND, B.C.    |
| <b>Software Developer</b>                                                                                                                                                                                                                                                                                                                | 09/2022 – 07/2025 |
| <ul style="list-style-type: none"><li>• Began as an intern developing an enterprise Java project based on the Cúram platform.</li><li>• Promoted to full-stack polyglot role building Angular SPAs, REST APIs and event-driven systems.</li></ul>                                                                                        |                   |
| The University of British Columbia                                                                                                                                                                                                                                                                                                       | VANCOUVER, B.C.   |
| <b>Graduate Teaching Assistant</b>                                                                                                                                                                                                                                                                                                       | 09/2018 – 04/2021 |
| <ul style="list-style-type: none"><li>• Planned and facilitated delivery of course materials for undergraduate courses.</li><li>• Communicated with students to identify research interests and address knowledge gaps.</li><li>• Evaluated performance on assignments, essays and exams.</li></ul>                                      |                   |

## Projects

- Meta-λ
- Continuation-passing interpreter for extended minimal subset of a call-by-name variation on Scheme.
  - Implementation of a reflective untyped lambda calculus, which makes apparent some connections between modal logics, metaprogramming, recursion theory, and continuation handling constructs.
  - Tech: #lang Racket (a Scheme variant)
- Guiser
- Web application using generative A.I. to automate creation and publication of social media content.
  - Enables a user to manage multiple personas, create topical content under a given guise with the assistance of large language models, then post that content to associated profiles on various platforms.
  - Tech: TypeScript, JavaScript, MERN (MongoDB, Express, React, Node.js), OAuth, LLM APIs
- search-factory
- Library of higher-order functions that produce graph search functions to client specifications.
  - Provides a unified framework for specifying symbolic A.I. problem-solving strategies.
  - Tech: #lang Racket (a Scheme variant)
- nice-org-html
- Open source Emacs package that generates customizable static websites from hyperlinked Org Mode markup files and Emacs themes, with defaults optimized for code documentation. 600+ users.
  - Tech: Emacs Lisp, JavaScript, CSS
- GNU Emacs (Contributor)
- Upgraded Emacs terminal display to support 24-bit colour on MS Windows 10+.
  - Tech: C, Emacs Lisp, Windows Console API, ANSI escape sequences

## Education

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|------------------------------------------------|------------------|
| University of British Columbia                 | VANCOUVER, B.C.  |
| <b>Bachelor of Computer Science</b>            | 2021 – 2024      |
| <b>M.A. (Ths.) Philosophy</b>                  | 2018 – 2021      |
| University of Lethbridge                       | LETHBRIDGE, A.B. |
| <b>B.A. (Hon.) Philosophy and Anthropology</b> | 2012 – 2017      |

## Technologies

**Languages:** Java, JavaScript, TypeScript, C#, C, Scheme, Lisp, HTML, CSS, UIM (Cúram DSL), SQL,  $\text{\LaTeX}$   
**Frameworks/Libraries:** Cúram, Node, Express, React, Redux, Angular, NgRx, Selenium, OpenTelemetry  
**Tools:** Emacs, Eclipse, IntelliJ, Visual Studio, Git, Linux, MacOS, Windows, CI/CD on Azure DevOps, GitHub